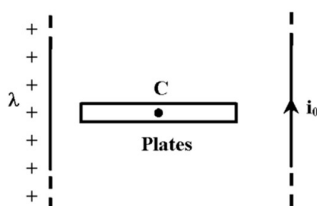
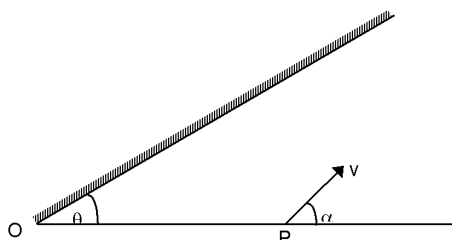


- Given two radioactive elements, a parent having a half life T_1 and a daughter product having a half life T_2 . If $T_1 > T_2$ and if initially there is only the parent element present, how long will it take in terms of T_2 until these elements are within 1 percent of their equilibrium value.
(A) $6.64 T_2$ (B) T_2 (C) $3T_2$ (D) $2 T_2$
- A uniform infinite thread of charge with linear charge density λ and another infinitely long parallel conducting wire lie in same plane with distance d between them. Current i_c flows in conducting wire as shown in the figure. A photoelectric plate PQ with work function ϕ is placed perpendicular to the wires with C as midpoint on the plate between the wires.



Determine the frequency of light to be radiated on the surface, for which fastest photoelectron ejected from C, normal to plate goes undeviated.

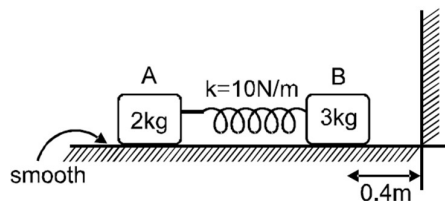
- (A) $\frac{1}{h} \left[m \left(\frac{\lambda}{\mu_0 i_0 \epsilon_0} \right)^2 + \phi \right]$ (B) $\frac{1}{h} \left[\frac{1}{2} m \left(\frac{\lambda}{\mu_0 i_0 \epsilon_0} \right)^2 + \phi \right]$
- (C) $\frac{2}{h} \left[\frac{1}{2} m \left(\frac{\lambda}{\mu_0 i_0 \epsilon_0} \right)^2 + \phi \right]$ (D) None of these
- A plane mirror is inclined at an angle $\theta = 60^\circ$ with horizontal surface. A particle is projected from point P (see figure) at $t = 0$ with a velocity v at an angle α with horizontal. The image of the particle is observed from the frame of the particle projected. Assuming the particle does not collide the mirror. Find the time when image will come momentarily at rest with respect to particle.



- (A) $\frac{u \cos \alpha (\tan \alpha - \tan \theta)}{g}$ (B) $\frac{u \cos \alpha (\tan \alpha + \tan \theta)}{g}$
- (C) $\frac{u (\tan \alpha - \tan \theta)}{g}$ (D) $\frac{u \sin \alpha (\tan \alpha + \tan \theta)}{g}$

4. If the wavelength of the incident radiation is increased from 4000 Å to 4002 Å in photoelectric effect. Find the corresponding change in the stopping potential.
 (A) 15.6 mV (B) -2.6 mV (C) -56 mV (D) -1.56 mV
5. A person A is fixed on the edge of a horizontal " Merry go-round" that has a radius of 6m and is rotating steadily. Another person B is standing still on the ground at a point that is 12m from the centre of the " Merry-go-round". Assume their line of sight is horizontal. At a particular instant, B observes A moving directly towards him with a speed of 1m/s and observes that there is no component of velocity perpendicular to the line joining them. At the same moment A observes B to be moving at a speed $\sqrt{n}$ m/s where n is an integer. At the same moment A observes that the rate at which the distance between them is decreasing is $\sqrt{m}$ m/s where m is an integer. At the same moment A observes B's velocity perpendicular to the line joining them to be $\sqrt{p}$ m/s where p is an integer. Choose the CORRECT option(s).
 (A) $m = 1$ (B) $n = 4$
 (C) $p = 3$ (D) Acceleration of A as observed by B is a constant
6. A particle moves in XY Plane according to the relations $x = kt$ and $y = kt(1 - pt)$ where k and p are positive constants and t is time. Therefore,
 (A) the trajectory of the particle is parabola
 (B) the particle has constant velocity along X axis
 (C) the force acting on the particle remains in the same direction even if both k and p are negative constants.
 (D) the particle has a constant acceleration along -Y axis
7. Velocity of a particle of mass 2 kg changes from $\vec{V}_1 = -2\hat{i} - 2\hat{j}$ m/s to $\vec{V}_2 = \hat{i} - \hat{j}$ m/s after colliding with a plane friction less surface. Choose the correct option(s) regarding this collision :
 (A) The angle made by the plane surface with the positive x-axis is $90^\circ + \tan^{-1}\left(\frac{1}{3}\right)$.
 (B) The angle made by the plane surface with the positive x-axis is $\tan^{-1}\left(\frac{1}{3}\right)$.
 (C) The direction of change in momentum of the particle makes an angle $\tan^{-1}\left(\frac{1}{3}\right)$ with the positive xaxis.
 (D) The direction of change in momentum of the particle makes an angle $90^\circ + \tan^{-1}\left(\frac{1}{3}\right)$ with the plane surface.
8. A particle is projected with a speed 100 m/s from an inclined plane at angle of 15° from the incline. The inclined plane has an angle of inclination 30° . The correct statements are :
 (A) The angle made by projectile with the plane at the time of lading is 37.8°
 (B) Range along the plane is about 488m
 (C) Horizontal displacement has a magnitude 422.6m
 (D) Vertical displacement has a magnitude 244m

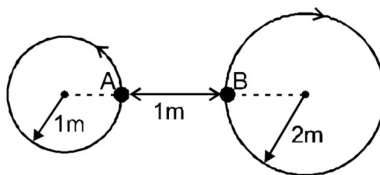
9. Initially distance between two small blocks A and B is 1m and natural length of spring is 2m. Now the system is released. If B hits the vertical wall elastically then : (All surfaces are smooth)



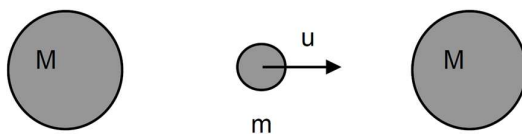
- (A) Maximum extension in spring during complete motion is 20cm.
- (B) Maximum speed of block B during complete motion is $\frac{2}{\sqrt{3}}$ m/sec.
- (C) At maximum extension speed of block A is $\frac{4\sqrt{3}}{5}$ m/sec.
- (D) Impulse by vertical wall on block B is $4\sqrt{3}$ N. sec.
10. A particle moves with constant speed v along a regular hexagon ABCDEF in the same order. Then the magnitude of the average velocity for its motion from A to :
- (A) F is $\frac{v}{5}$ (B) D is $\frac{v}{3}$ (C) C is $\frac{v\sqrt{3}}{2}$ (D) B is v .
11. A particle moving in a straight line has acceleration given by $a = -k\sqrt{v}$ where k is a positive constant and v is instantaneous velocity. Let v_0 be the initial velocity, t be the total time of motion and v_a be the average velocity in time t . Let s be total distance travelled. Choose the correct options.
- (A) $t = \frac{2\sqrt{v_0}}{k}$ (B) $v_a = \frac{v_0}{3}$ (C) $s = \frac{2}{3} \frac{\sqrt{v_0}}{k}$ (D) $\frac{v_a}{v_0} = \frac{1}{2}$
12. A helicopter takes off along the vertical with an acceleration $a = 3 \text{ m/s}^2$ and zero initial velocity. In a certain time t_1 pilot switches off the engine. At the point of take off, the sound dies away in a time $t_2 = 30 \text{ sec}$. Speed of sound is 320 m/s. Choose the correct option(s).
- (A) Pilot switches off the engine at height $h = \frac{4200}{3} \text{ m}$
- (B) Pilot switches off the engine in time $t_1 = \frac{80}{3} \text{ sec}$
- (C) Pilot switches off the engine at height $h = \frac{3200}{3} \text{ m}$
- (D) Speed of helicopter at the moment when pilot switches off the engine is 80 m/s

Comprehension

Two particles are moving in different circles in same plane with different angular velocities as shown in figure. At $t = 0$, initial positions of particles A and B are shown by dots on the respective circles. Initial distance between particles is 1m. Particle A move anticlockwise in the first circle whereas B moves clockwise in the second circle. Angle described (rotated) by A and B in time 't' are $\theta_A = \left(\frac{\pi}{2}t\right)$ and $\theta_B = (\pi t)$ respectively. Here θ is in radian and t is in second. Radius of each circle is shown in diagram.

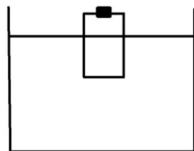


13. At time $t = 1$ sec, the magnitude of acceleration of A with respect to B is
 (A) $\frac{\pi^2}{4}\sqrt{65}\frac{\text{m}}{\text{sec}^2}$ (B) $\frac{\pi^2}{2}\sqrt{7}\frac{\text{m}}{\text{sec}^2}$ (C) $\frac{\pi^2}{3}\sqrt{15}\frac{\text{m}}{\text{sec}^2}$ (D) $\frac{\pi^2}{4}\sqrt{7}\frac{\text{m}}{\text{sec}^2}$
14. At time $t = 2$ second, the angular velocity of the particle A with respect to the particle B is
 (A) 5π rad/sec (B) $\frac{3\pi}{2}$ rad/sec (C) $\frac{2\pi}{3}$ rad/sec (D) $\frac{5\pi}{6}$ rad/sec
15. A stone is dropped from the top of building and at the same time a second stone is thrown vertically upward from the bottom of the building with a speed of 20 ms^{-1} . They pass each other 3 seconds later. Find the height of the building.
16. Two identical uniform spheres of mass M are at rest and another uniform sphere of mass m is moving along their line of centers between them with speed u . After second collision middle sphere move with velocity $0.36u$. If spheres are perfectly elastic then calculate.



17. The maximum possible acceleration of a train moving on straight track is 10m/s^2 and maximum possible retardation is 5m/s^2 . Find the minimum time in which the train can complete a journey of 135m. The maximum speed that train can achieve is 135m/s.
18. Moment of inertia of a disc of mass $M = 2 \pm 0.02$ kg, inner radius $a = 5 \pm 0.05$ cm, outer radius $b = 7 \pm 0.07$ cm, with respect to an axis passing through geometrical centre of the disc and perpendicular to the plane of disc is given by $(A_0 \pm B_0 \times 10^{-2})\text{ kg} - \text{cm}^2$. Find the value of $\frac{B_0}{A_0}$.
19. A thin uniform rod of mass 0.25 kg and length 1.20 m is smoothly hinged about a point passing through its centre of mass. Its angular momentum as a function of time t is given by $L = \frac{2}{3}t^3 + 2t^2$ (where t is in sec. are L is in $\text{kg m}^2/\text{sec}$). Angular acceleration of the rod at $t = \frac{3}{2}$ sec. is $50X$ (in rad/sec^2) then find 'X'.

20. A uniform wooden box of base area 10 cm^2 , height 10 cm and specific gravity 0.6 floats in water while a coin of mass 20 g and specific gravity 2 lies above it. The area of the base of container is 100 cm^2 . Now, the coin is picked up from the block and put directly into the container. Find the change in level of water in the container in mm.



ANSWER KEY

1.	(A)	2.	(B)	3.	(A)	4.	(D)	5.	(ABC)
6.	(ABCD)	7.	(AC)	8.	(BCD)	9.	(ACD)	10.	(ACD)
11.	(AB)	12.	(BCD)	13.	(A)	14.	(D)		
15.	60.00 m	16.	4	17.	09.00 sec	18.	3	19.	7
20.	1								